# 9. Dependencies

## 9.1 Core Requirements

| Package | Purpose |
|---|---|
| streamlit >= 1.35.0 | Web UI framework — multipage app, session state, widgets |
| numpy >= 1.26.0 | Numerical arrays used throughout backend |
| pandas >= 2.2.0 | DataFrame operations, CSV/Excel parsing |
| scikit-learn >= 1.5.0 | All sklearn models, preprocessing, metrics, CV |
| joblib >= 1.3.0 | Model serialisation and caching |
| optuna >= 3.6.0 | TPE hyperparameter search (primary autotune method) |
| matplotlib >= 3.8.0 | All 9 plot functions |
| seaborn >= 0.13.0 | Correlation heatmap styling |
| plotly >= 5.22.0 | Interactive charts on comparison pages |
| scipy >= 1.13.0 | Skewness, kurtosis, loguniform for RandomizedSearchCV |
| python-docx >= 1.1.0 | Word report generation |
| weasyprint >= 62.0 | HTML → PDF conversion (requires GTK3 system libs) |
| jinja2 >= 3.1.0 | HTML template rendering for PDF pipeline |
| python-dotenv >= 1.0.0 | .env config loading |
| Pillow >= 10.3.0 | Image handling — embed plots into docx/PDF |
| openpyxl >= 3.1.0 | Excel file upload support (.xlsx) |

## 9.2 Optional Dependencies

| Package | Notes |
|---|---|
| lightgbm >= 4.3.0 | LightGBM model (pip install lightgbm). Loaded conditionally. |
| catboost >= 1.2.0 | CatBoost model (pip install catboost). Loaded conditionally. |
| xgboost >= 2.0.0 | Native XGBoost API. Not required — HistGradientBoosting covers most use cases. |

## 9.3 System Dependencies (WeasyPrint)

WeasyPrint requires GTK3 system libraries that pip cannot install:

| Platform | Command |
|---|---|
| Ubuntu / Debian (WSL) | sudo apt install libpango-1.0-0 libpangoft2-1.0-0 libgdk-pixbuf2.0-0 libffi-dev |
| Windows (native) | Install GTK3 runtime from: github.com/tschoonj/GTK-for-Windows-Runtime-Environment-Installer. Enable 'Add to PATH' during install. |
| macOS | brew install pango gdk-pixbuf libffi |